Hammad Ahmad

Data Scientist | Predictive Modeling · Statistical ML · Python

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LinkedIn | GitHub | Portfolio | ORCID

PROFILE

Data scientist with an MSc (Merit) in Applied AI and a first-author peer-reviewed publication benchmarking machine learning methods (Springer). I build and evaluate predictive models on real-world clinical and text data, from exploratory analysis through ensemble modelling to REST deployment, and work fluently across the Python data stack.

TECHNICAL SKILLS

Machine Learning & Statistics : Predictive modeling, classification, ensemble methods (Random Forest, XGBoost, LightGBM, CatBoost), class-imbalance handling (SMOTE, ADASYN, over-sampling), scikit-learn, PyTorch, TensorFlow, MLflow

Data Analysis : pandas, NumPy, SQL, feature engineering, model evaluation (ROC-AUC, sensitivity, specificity), Matplotlib, Seaborn

NLP & LLMs : LLMs, Retrieval-Augmented Generation (RAG), Sentence Transformers, RAGAS evaluation, Neo4j knowledge graphs, Ollama

Programming & Tools : Python, C++, FastAPI, Flask, REST APIs, PostgreSQL, Docker, Git, Linux

EXPERIENCE

Outlyst — Oct 2025 - Mar 2026
AI / Machine Learning Engineer — Leeds, UK (Remote)

- Built backend dialogue-flow logic and workflow automation for an AI voice-agent system (Retell AI, FastAPI), integrating LLM-based dialogue with production infrastructure.
- Developed asynchronous backend services and lead-tracking automation for an AI-driven outbound communication platform.

University of Bradford — Jan 2025 - Sep 2025
Research Assistant, Machine Learning & LLMs — Bradford, UK

- Built FinLaw-UK, a RAG system pairing Mistral 7B (served locally via Ollama) with a Neo4j knowledge graph for UK financial-regulation question answering, grounded in FCA, PRA, FRC and statutory sources.
- Engineered the retrieval pipeline: clause-level segmentation, Sentence Transformer embeddings, cross-encoder re-ranking, and Neo4j citation checks that flag hallucinated references.
- Evaluated on a 110-item legal QA set with RAGAS and custom metrics: 0.82 source accuracy and 0.81 citation quality, improving on a standalone LLM baseline.

COMSATS University Islamabad — Jul 2023 - Jul 2024
Research Assistant, Data Science — Islamabad, Pakistan

- Benchmarked 11 classifiers (Random Forest, XGBoost, LightGBM and others) for diabetes risk on a 253,680-record CDC BRFSS dataset; Random Forest reached 93.2% accuracy and 0.99 AUC after correcting severe class imbalance with random over-sampling.
- Analysed 20+ health indicators, identifying age, general health, BMI, blood pressure and income as the strongest diabetes correlates.
- Deployed the model as a risk-screening web app (React, Flask REST API) returning personalised risk probabilities.

PROJECTS

Jobzyl: Multi-Platform Job Aggregator — jobzyl.com
Full-Stack Architecture, Data Pipelines, Responsive UI

- Built a full-stack platform that ingests real-time job listings from multiple sources (LinkedIn, Indeed, Google Jobs) into a single searchable portal; designed the parsing pipelines and responsive UI. Live and actively maintained.

EDUCATION

University of Bradford — Sep 2024 - Sep 2025
MSc, Applied Artificial Intelligence and Data Analytics (Merit) — Bradford, UK

- **Dissertation**: FinLaw-UK: A Graph-Augmented Retrieval Chatbot for Reliable and Transparent UK Financial Regulation.

COMSATS University Islamabad — Sep 2020 - Jul 2024
BS, Bioinformatics — Islamabad, Pakistan

PUBLICATIONS

Ahmad, H., Khan, M.U., & Azam, M. (2024). Comparative Analysis of Machine Learning Methods for Enhancing Sleep Efficiency and Prediction. *ICSMAI 2024* (Springer). DOI: 10.1007/978-3-031-66854-8_1

LANGUAGES: English (IELTS 7.0, CEFR C1) | Urdu (Native) | German (A1.2)